PLANRS ACADEMY MASTER COURSE

Grade 4 Engineering | Robotics Series

The Magic Switch

Making Motors Dance!

SESSION 1: H-BRIDGE CIRCUIT BASICS

Welcome, Young Engineers!

Today we are going to learn how to control a motor's "mood." Have you ever wondered why your toy car moves forward when you push one button, but zooms backward when you push another?

What is a Motor?

Motors are amazing devices that turn electricity into spinning motion! They are like magical spinners hiding inside objects you use every day:

- Ceiling Fans: They spin to keep you cool.
- Washing Machines: They spin to clean your clothes.
- Toy Cars: Motors make their wheels zoom.

The Big Question: Can the same motor and same battery go in both directions? Let's find out!

Meet the H-Bridge

The H-Bridge is a special circuit that looks like the letter 'H'! It is our "Magic Control Panel."

HOW IT WORKS

The H-Bridge uses 4 switches that work together as a team to control which way electricity flows to your motor.

Electricity is like Water!

To change the direction a motor spins, we have to change the direction the electricity flows. We can look at it like water pipelines:

💧 If water flows left-to-right, the wheel spins **clockwise (Forward)**.

💧 If water flows right-to-left, the wheel spins **counter-clockwise (Backward)**.

[Visual Note: A diagram resembling the letter 'H' showing a DC Motor in the center link, connected to four switches S1, S2, S3, and S4.]

The Dance Steps: Steering the Current

By opening and closing opposite switches, we control the direction of the electricity!

Dance Step 1: Spin Forward!

When we turn ON Switch 1 and Switch 4 (while leaving Switch 2 and Switch 3 OFF), the electric current marches from the left side, travels rightwards through the motor, and goes back to the battery.

Result: The motor spins Forward!

Dance Step 2: Spin Backward!

When we turn ON Switch 2 and Switch 3 (while leaving Switch 1 and Switch 4 OFF), the path reverses. The current enters the motor from the opposite side and flows leftwards instead.

Result: The motor spins Backward!

The Engineering Safety Rule

When working with H-Bridge systems, there is one critical rule that every Robotics Engineer must memorize to keep their circuit board safe.

🛑 THE SHORT-CIRCUIT DANGER!

NEVER turn on Switch 1 and Switch 2 at the same time, or Switch 3 and Switch 4 at the same time.

If you close them simultaneously, the electricity will bypass the motor completely and rush straight from the power terminal down to the ground. This causes a **Short Circuit**, which makes the wires extremely hot and can ruin your battery or melt your components!

[Visual Note: Arrow indicator showing a short-circuit bypass route down a vertical leg when S1 and S2 are closed together.]

The Construction Blueprint

Let's build a real, working H-Bridge using simple items in the lab! Follow these instructions step-by-step:

Step 1: Lay the Foundation
Draw a large letter 'H' outline on a clean piece of cardboard using a dark marker.

Step 2: Place the Actor
Fix your small DC toy motor right onto the middle horizontal link of the 'H' diagram.

Step 3: Route the Wires
Secure metallic paths along the paths of the letter 'H' to connect your motor terminals to the switches. Make sure you leave open gaps on the side rails to act as your four manual switches.

Step 4: Test Your Magic!
Connect a 9V battery at the top and bottom hubs. Flip the switches one way. Does the motor spin forward? Now flip them the other way. Did it spin backward?

The "Wiper" Challenge

Now that your motor is "dancing," let's give it a job!

Your Goal: Use your motor and H-Bridge to move a small cardboard 'wiper' back and forth like a car's windshield wiper.

MEASUREMENT TABLE

Try different objects on your motor. Does it still spin both ways?

Object	Forward Spin? (Yes/No)	Backward Spin? (Yes/No)
Small Tape Piece		
Cardboard Wiper		

The Circuit Captain Quiz

Check if you have earned your Engineering Badge! Circle the correct answer:

1. What happens if we flip the battery wires connected to a motor?

a) The motor stops
b) The motor spins in the opposite direction
c) The motor explodes

2. How many switches are in a basic H-Bridge team?

a) Two
b) Four
c) One hundred

3. Which everyday item uses an H-Bridge to clean?

a) A ceiling fan
b) A washing machine
c) A simple table lamp

Lesson Recap

You have mastered the foundational concepts of motor control circuits! Let's review the main takeaways from our lab session:

- ⚙️ **Motor Direction:** Changing the direction that electric current travels reverses the motor spin.
- ⚙️ **H-Bridge Layout:** A circuit configuration with four independent switches grouped in pairs to route current.
- ⚙️ **Safety Priority:** Avoiding unsafe, adjacent switch combinations that create short circuits and cause thermal strain.
- ⚙️ **Practical Automation:** H-bridge mechanisms run automated directional movements like windshield wipers and vehicle powertrains.

Circuit Captain Certified!

Keep testing, keep tinkering, and never stop building!